



FINAL REPORT

Building Semantic Corpus of Scholarly Publications on Global warming and Temperature rise

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I. ABSTRACT

Semantic corpora development for metadata and encyclopedia

The increasing availability of scientific publications has created a need for efficient methods to organize, retrieve, and analyse research information in a structured and reproducible manner. **Semantic corpora** provide an effective solution by combining **literature retrieval** with **metadata organization**, enabling researchers to efficiently manage large collections of scientific publications and support evidence-based research. The present internship focused on developing a semantic corpus for the research topic "**Global Warming and Temperature Rise**" using a systematic computational workflow.

The work involved formulating a research query, retrieving relevant scientific records from scholarly repositories, organizing bibliographic metadata into a standardized directory structure, generating review tables, and creating an interactive review environment to facilitate literature screening using tools like **pygetpapers**, **Prisma 2020** and **HTML**. Throughout the project, emphasis was placed on reproducibility by maintaining consistent file organization, documenting workflow stages, and preserving metadata required for future reuse. The internship also provided practical experience in using computational tools and version control systems to support collaborative research and transparent data management.

The resulting semantic corpus represents a structured collection of climate-related scientific literature that can serve as a foundation for systematic reviews, bibliometric studies, semantic search, and natural language processing applications.

Keywords: Semantic corpus, pygetpapers, Prisma 2020, HTML, scientific literature retrieval, metadata management, reproducible research.

II. INTRODUCTION

Scientific literature retrieval involves identifying, collecting, and organizing scholarly publications relevant to a research question. Traditionally, literature retrieval has relied on keyword-based searches within bibliographic databases and digital repositories. Modern literature retrieval increasingly incorporates metadata-based approaches, semantic indexing, citation analysis, and machine-assisted filtering to improve both precision and recall. Metadata—including titles, abstracts, authors, publication years, journals, Digital Object Identifiers (DOIs), keywords, and repository identifiers—provides structured information that facilitates efficient organization and retrieval.

For systematic reviews and evidence synthesis, literature retrieval must also ensure transparency and reproducibility. Recording search queries, retrieval dates, repository sources, and inclusion criteria enables other researchers to reproduce the search process and verify the resulting dataset.

During this internship, literature retrieval formed the initial stage of semantic corpus development. Retrieved publications were organized systematically to create a reusable and reproducible dataset for further analysis.

Semantic Corpora

A corpus is a structured collection of documents assembled for a specific purpose. In computational research, corpora often consist of text documents accompanied by metadata and annotations. A **semantic corpus** extends this concept by incorporating meaningful relationships between documents, entities, concepts, and metadata, allowing richer exploration and analysis than a simple document repository.

Semantic corpora provide several advantages:

- Standardized organization of scientific literature.
- Improved retrieval through metadata and semantic relationships.
- Support for reproducible research workflows.
- Compatibility with natural language processing and machine learning techniques.
- Facilitation of systematic reviews and evidence synthesis.
- Integration with knowledge graphs and semantic web technologies.

Reproducible Research and FAIR Principles

Scientific reproducibility is a fundamental principle of modern research. Reproducible workflows allow independent researchers to obtain comparable results by following the same methodology using the same data and computational procedures. In literature-based research,

reproducibility requires clear documentation of search strategies, repository sources, screening criteria, and data organization methods.

The **FAIR Data Principles**—Findable, Accessible, Interoperable, and Reusable—provide internationally recognized guidelines for research data management. Applying these principles to semantic corpus construction improves transparency, facilitates collaboration, and enhances the long-term value of research datasets.

Within this internship, reproducibility was emphasized through standardized directory structures, consistent metadata organization, documented workflows, and structured review processes. These practices align with FAIR principles by making the corpus easier to understand, reproduce, and extend.

III. OBJECTIVES

The internship was designed around the following objectives:

1. To understand the principles of semantic corpus construction and scientific literature organization.
2. To formulate an effective search query related to **Global Warming and Temperature Rise**.
3. To retrieve relevant scientific publications from literature repositories using standardized search strategies.
4. To organize retrieved publications into a structured directory hierarchy.
5. To collect and manage publication metadata in a consistent and reproducible format.
6. To perform literature screening using predefined review criteria.
7. To prepare review tables for efficient assessment of retrieved articles.
8. To generate an interactive HTML-based review interface for literature exploration.
9. To maintain reproducibility through standardized workflows and documentation.
10. To develop a reusable semantic corpus that can support future research, systematic reviews, and computational analyses.

IV. DETAILED OVERVIEW OF WORK:

4.1. Flow chart:

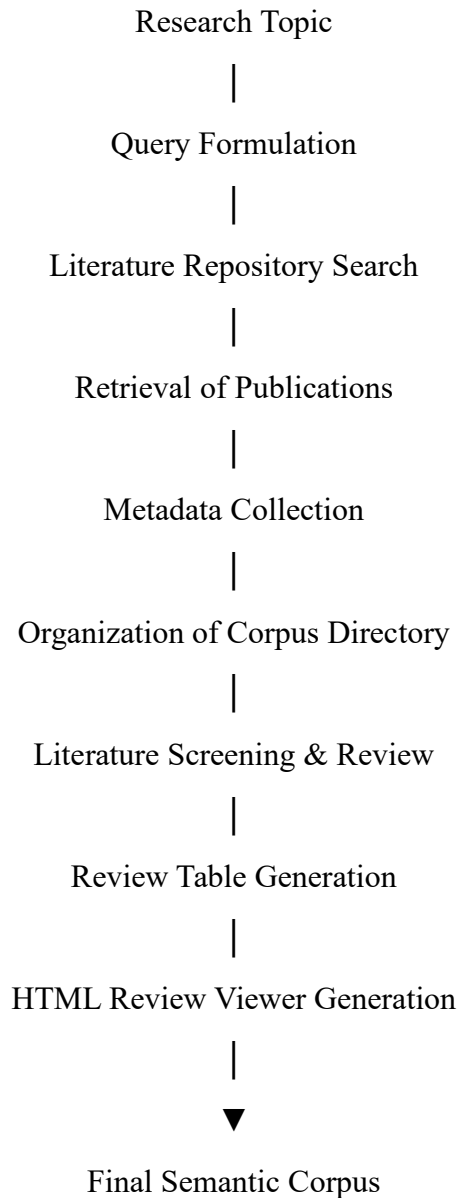


Fig.4.1. Overview of the project workflow

4.2 Selection of the Research Query

The first step in the corpus development process involved selecting an appropriate research topic. The query **"Global Warming and Temperature Rise"** was chosen due to its scientific significance and the availability of a substantial body of peer-reviewed literature.

This topic spans multiple disciplines, including:

- Climate Science
- Environmental Science
- Atmospheric Science
- Ecology
- Agriculture
- Public Health
- Renewable Energy
- Environmental Policy

Because of its interdisciplinary nature, the query provides a suitable case study for demonstrating semantic corpus construction and literature organization.

The search query was standardized and consistently used throughout the workflow to maintain reproducibility.

```
In [ ]: from pathlib import Path

QUERY_NAME = "global_warming"
QUERY_STRING = (
    '("global warming" OR "climate change" '
    'OR "greenhouse effect" '
    'OR "global temperature rise")'
)
LIMIT = 50
FORMATS = ["xml"]
REPOSITORY = "europe_pmc"
NOTES = ""
REVISION_OF = None # e.g. "aqi_india_pilot" when refining a prior run

OUTPUT_DIR = Path("/content/queries", QUERY_NAME)
OUTPUT_DIR
```

Fig.4.2 Overview of the pipeline with query

4.3 Literature Retrieval

After defining the research query, relevant scientific publications were retrieved from academic literature repositories using the semantic corpus workflow.

The retrieval process aimed to identify peer-reviewed publications relevant to global warming and temperature rise while preserving publication metadata required for subsequent stages of corpus construction.

The retrieved information included:

- Article title
- Authors

- Publication year
- Journal
- Abstract
- DOI
- Repository information
- Keywords
- Publication identifiers

Maintaining complete metadata ensured that publications could be efficiently organized and reviewed later in the workflow.

4.4 Corpus Directory Organization

A standardized directory structure was created to organize all resources associated with the corpus. Maintaining a consistent folder hierarchy improves reproducibility and simplifies navigation.

The corpus is created at

https://github.com/semanticClimate/semantic_corpus/tree/shreya/corpora

4.5 Metadata Collection

Metadata collection represented one of the most important stages of corpus construction.

Rather than storing only PDF files, each publication was accompanied by structured metadata describing its bibliographic information.

The metadata fields included:

Table.4.3. List of the metadata field and description

Field	Description
Title	Research article title
Authors	List of authors
Year	Year of publication
Journal	Publishing journal
DOI	Digital Object Identifier
Abstract	Article summary
Keywords	Indexing terms
Repository	Source repository

Organizing publications using metadata provides several advantages:

- Efficient searching
- Filtering by publication year
- Sorting by author
- Grouping publications
- Integration with computational tools
- Future knowledge graph construction

Metadata files were maintained in structured formats that could be easily processed by automated scripts.

The metadata is available on GitHub at

https://github.com/semanticClimate/semantic_corpus/tree/shreya/corpora/shreya_project

4.6 Literature Screening

Following literature retrieval, publications underwent a screening process to determine their relevance to the selected research topic.

Screening involved examining publication titles, abstracts, and associated metadata.

The objective of screening was to remove publications that were unrelated or only marginally relevant to the research question while retaining studies directly addressing global warming and temperature rise.

The screening workflow consisted of:

1. Initial retrieval of publications.
2. Duplicate identification.
3. Title screening.
4. Abstract screening.
5. Eligibility assessment.
6. Selection for review.

A simplified representation is shown below.

Link:

https://github.com/semanticClimate/semantic_corpus/blob/shreya/corpora/shreya_project/prisma_diagram.png

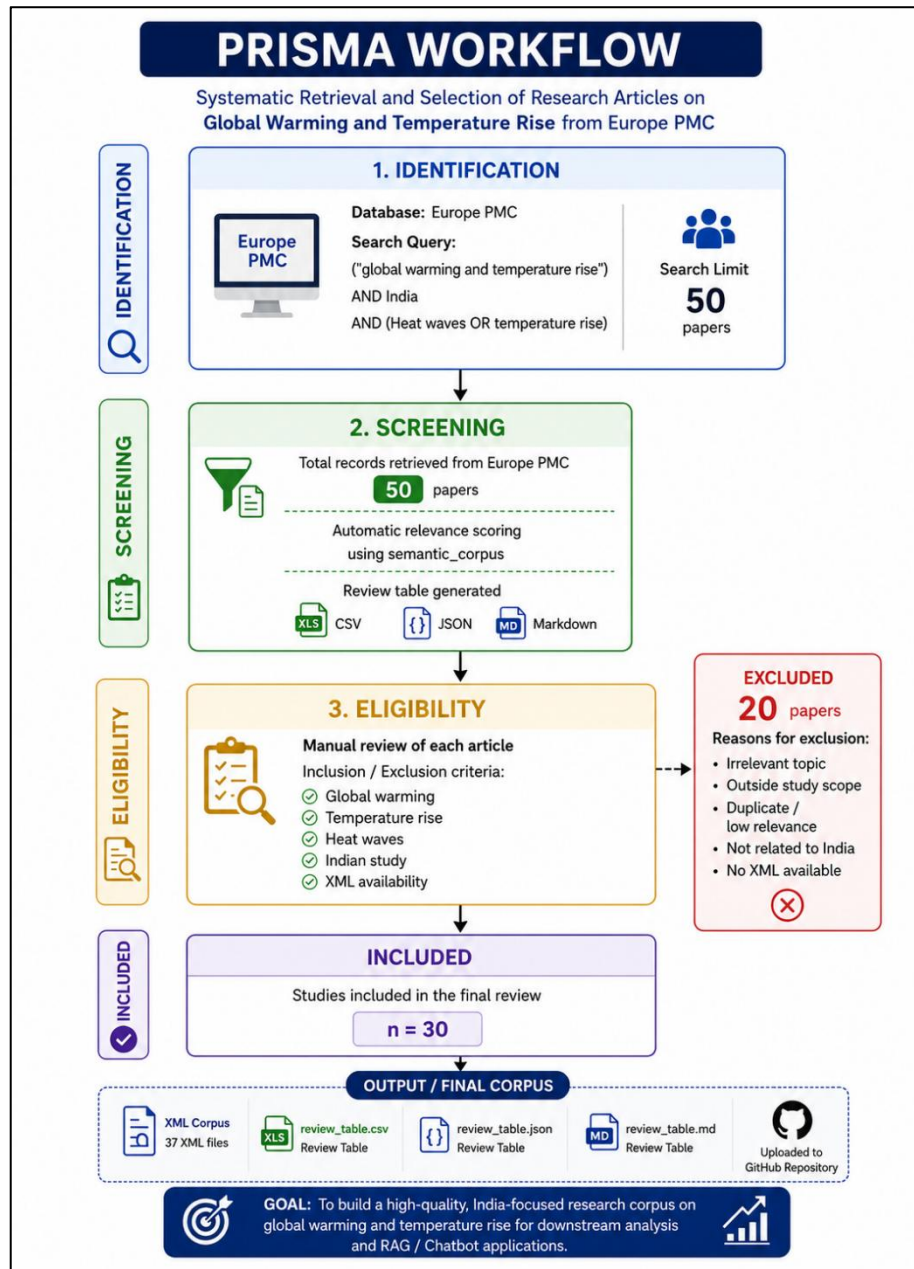


Figure 4.4. Simplified literature screening workflow using Prisma 2020

4.8 Review Table Generation

To facilitate efficient evaluation of retrieved publications, review tables were generated during the corpus construction process.

The review table served as a structured summary of all publications included within the corpus.

Each record typically contained:

- Article title
- Authors

- Publication year
- Journal
- Repository
- Review status
- Relevance
- Notes

The review table enabled systematic comparison between publications and simplified literature assessment.

An HTML-based review interface was subsequently generated from these review tables to improve accessibility and navigation.

The review_table is available at

https://github.com/semanticClimate/semantic_corpus/blob/shreya/corpora/shreya_project/review_table.csv

Table.4.5. Review_table containing pmcid, pmid and title of paper

review_status	score	paper_id	pmcid	pmid	doi	title
include	4	europe_pmc_PMC11189284	PMC11189284	38911103	10.4103/ijem.ijem_473_23	Global Warming and Endocrinology: The Hyderabad Declaration
include	4	europe_pmc_PMC12890902	PMC12890902	41632847	10.1073/pnas.2502873123	Amplified warming in tropical and subtropical cities under 2 °C
include	4	europe_pmc_PMC12891482	PMC12891482	41545504	10.1038/s41598-026-36299-3	Emerging heat stress patterns across India under future climate
include	4	europe_pmc_PMC12972134	PMC12972134	41673178	10.1038/s41598-026-38289-x	Intensification and spatial shifts of heatwave hotspot across India
exclude	4	europe_pmc_PMC13286149	PMC13286149	42329980	10.1371/journal.pone.0351740	Cold stress during winter over North India: Patterns, trends, and
exclude	3	europe_pmc_PMC12891732	PMC12891732	41545467	10.1038/s41598-026-35482-w	Near future projections of tropical cyclone tracks over the Bay

Table.4.6. Review_table containing the authors

publication_date	journal	authors
2024-03-01		Bhattacharya S; Sahay R; Afsana F; Sheikh A; Widanage NM; Maskey R; Naseri MW; Murad M; Harikumar KVS; Selim S; Aa
2026-02-03		Berk S; Joshi MM; Goodess CM; Nowack P
2026-01-16		Molina MO; Soares PMM; Agarwal A; Trigo RM
2026-02-12		Banerjee S; Padmakumari B; Ramana MV
2026-06-22		Athira KS; Attada R; Komeravelli S
2026-01-16		Adsul R; Singh VK; Parekh A; Gnanaseelan C

Table.4.7. Review_table with information on xml, location and abstract

has_xml	has_pdf	query_name	query_string	location_terms
True	False	global_warming and temperature rise	("global warming" OR "temperature rise" OR "warm air" OR heat waves OR warming of earth) AND India	hyderabad
True	False	global_warming and temperature rise	("global warming" OR "temperature rise" OR "warm air" OR heat waves OR warming of earth) AND India	india
True	False	global_warming and temperature rise	("global warming" OR "temperature rise" OR "warm air" OR heat waves OR warming of earth) AND India	india
True	False	global_warming and temperature rise	("global warming" OR "temperature rise" OR "warm air" OR heat waves OR warming of earth) AND India	india, west bengal
True	False	global_warming and temperature rise	("global warming" OR "temperature rise" OR "warm air" OR heat waves OR warming of earth) AND India	india, uttar pradesh
True	False	global_warming and temperature rise	("global warming" OR "temperature rise" OR "warm air" OR heat waves OR warming of earth) AND India	india, indian

4.9 HTML Review Viewer

One of the notable components of the workflow was the generation of an HTML-based review viewer. This interactive interface allowed users to browse and inspect the review table through a web browser.

The HTML viewer improved usability by presenting publication metadata in an organized format and supporting easier navigation compared with raw JSON files.

Features of the review viewer included:

- Structured tabular presentation.
- Easy navigation between records.
- Display of publication metadata.
- Review status indicators.
- Improved readability for manual assessment.

Available on GitHub as

https://github.com/semanticClimate/semantic_corpus/blob/shreya/corpora/shreya_project/review_table_editor.html

Corpus review table										
50 papers · include 30 · exclude 20										
Connected to review server at https://silver-potato-6vv4g7v9vrggf5xxw-8765.app.github.dev . Click Read to open papers.										
Edit Status and Notes below. Style this page via <code>review_table.css</code> . Full papers load only when this page is opened via <code>review_viewer.py</code> serve (http:// URL, not a saved file).										
Status	Title	PMCID	Authors	Date	XML	PDF	Read	Abstract	Notes	
include	Global Warming and Endocrinology: The Hyderabad Declaration of the South Asian Federation of Endocrine Societies	PMC11189284	Bhattacharya S; Sahay R; Afsana F; Sheikh A; Widanage NM; Maskey R; Naseri MW; Murad M; Harikumar KVS; Selim S; Aamir AH; Muthukuda D; Parajuli N; Baheer MD; Latheef A; Nagendra L; Mondal S; Kamrul-Hasan ABM; Raza SA; Somasundaram N; Shrestha D; Anne B; Ramakrishnan S; Kalra S	2024-03-01	yes	yes	Read	Global warming and endocrine disorders are intertwined issues posing significant challenges. Greenhouse gases emanating from human activities drive global warming, leading to temperature rise and alte...		
include	Amplified warming in tropical and subtropical cities under 2 °C climate change.	PMC12890902	Berk S; Joshi MM; Goodess CM; Nowack P	2026-02-03	yes	yes	Read	Cities are often warmer than rural surroundings due to a phenomenon known as the urban heat island, which can be influenced by various factors, such as regional climate and land surface types. Under c...		
include	Emerging heat stress patterns across India under future climate scenarios.	PMC12891482	Molina MO; Soares PMM; Agarwal A; Trigo RM	2026-01-16	yes	yes	Read	This study investigates the climate change signal on mean and extreme temperatures in India, adding new insight by using the state-of-the-art CMIP6 projections to quantify the seasonal and spatial evo...		
include	Intensification and spatial shifts of heatwave hotspot across India under climate and ENSO influences with health risk assessment.	PMC12972134	Banerjee S; Padmakumari B; Ramana MV	2026-03-26	yes	yes	Read	Heatwaves (HWs) in India have intensified significantly over the past two decades, with projections of further increase in frequency and duration. While global warming mitigation efforts are advancing...		

Figure 4.8. HTML review table generated during corpus construction.

4.10 Semantic Corpus Organization

After screening and review, the selected publications were integrated into the final semantic corpus.

The completed corpus consisted of:

- Structured directory hierarchy.
- Downloaded literature.
- Metadata files.
- Review tables.
- HTML review interface.
- Supporting documentation.

The organization ensured that the corpus could be reused by other researchers without requiring reconstruction of the workflow.

V. OT-VIRARe TG and the Chilean National Committee

I actively participated in the demonstration of the chatbot representing the Climate Insight and the semantic corpus. This meeting called together the discussions on the IPCC chatbot and the semantic corpus with CODATA and other academicians across Argentina and Chile. Throughout the event, I contributed as a participant and enhanced my technical understanding of open-access data processing and global research ecosystem.

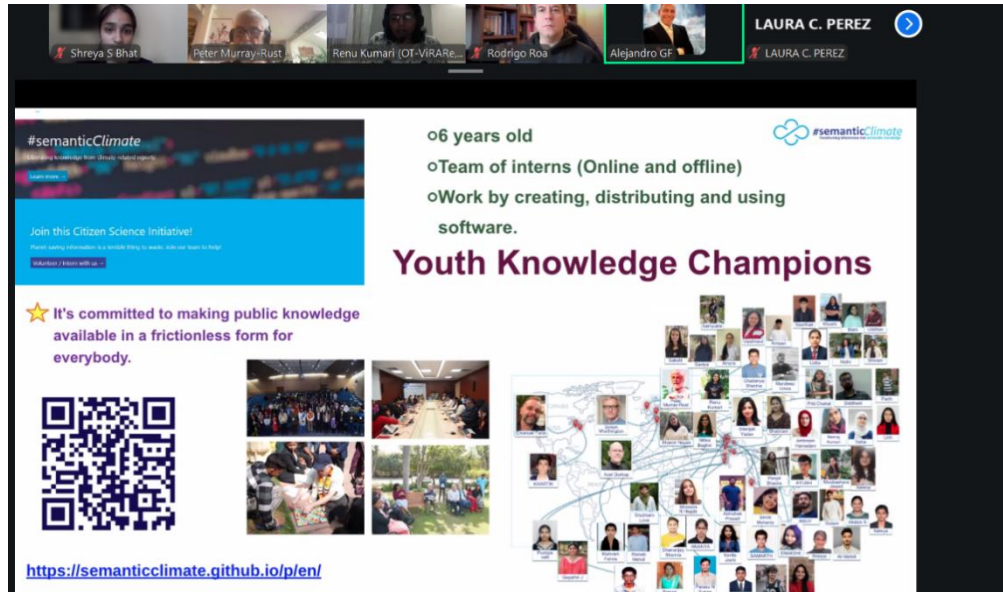


Fig.5.1. Introduction to semanticClimate

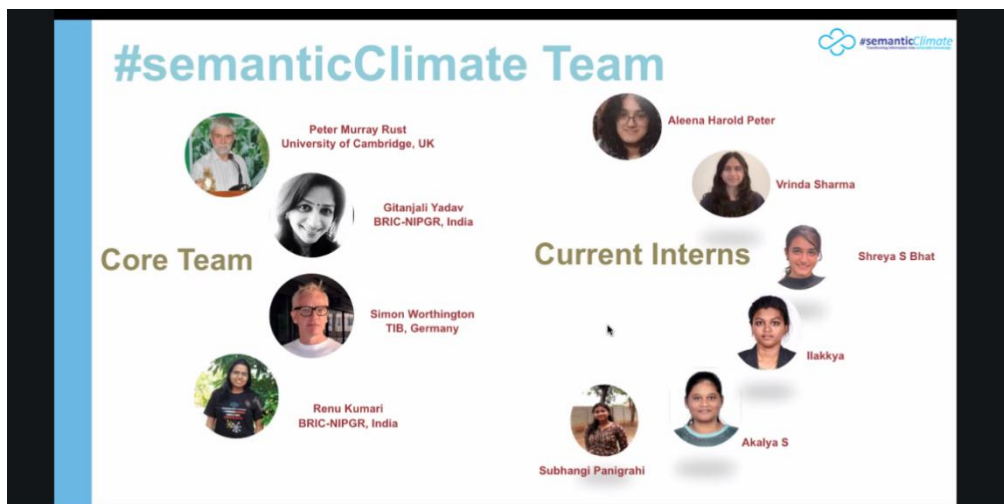


Fig.5.2. Current team of semanticClimate

VI. RESULTS

50 reports on global warming and temperature rise were collected from EuropePMC. Further, these papers were manually screened and 20 papers were excluded. All the papers were labelled as included or excluded in the corpus review table generated.

The papers were scored based on the relevance to the query generated. Those papers with score more than 5 are generally considered to be more relevant to the query.

Then, the included papers were stored as xml files in the GitHub corpora for the building of encyclopedia and chatbot. This can further be used by other scientists and enthusiasts for continued progress of semanticClimate.

The copy of the query is available on

https://github.com/semanticClimate/semantic_corpus/blob/shreya/corpora/shreya_project/Copy_of_corpus_query_review_colab.ipynb

On GitHub :

https://github.com/semanticClimate/semantic_corpus/tree/shreya/corpora/shreya_project/global_warming_and_temperature_rise/global_warming%20and%20temperature%20rise

 PMC111111018.xml	Add global_warming_and_temperature_rise folders
 PMC11189284.xml	Add global_warming_and_temperature_rise folders
 PMC11393388.xml	Add global_warming_and_temperature_rise folders
 PMC11873770.xml	Add global_warming_and_temperature_rise folders
 PMC11880391.xml	Add global_warming_and_temperature_rise folders
 PMC12167960.xml	Add global_warming_and_temperature_rise folders
 PMC12176879.xml	Add global_warming_and_temperature_rise folders
 PMC12181172.xml	Add global_warming_and_temperature_rise folders
 PMC12312808.xml	Add global_warming_and_temperature_rise folders
 PMC12422948.xml	Add global_warming_and_temperature_rise folders
 PMC12534177.xml	Add global_warming_and_temperature_rise folders
 PMC12540639.xml	Add global_warming_and_temperature_rise folders
 PMC12590180.xml	Add global_warming_and_temperature_rise folders

Fig.6.1. Reviewed pygetpapers as xml files

However, running the pipeline against the query gives the details of the papers in csv, json, CSS, HTML and md files.

csv :

https://github.com/semanticClimate/semantic_corpus/blob/shreya/corpora/shreya_project/global_warming_and_temperature_rise/global_warming%20and%20temperature%20rise/review/review_table.csv

json :

https://github.com/semanticClimate/semantic_corpus/blob/shreya/corpora/shreya_project/global_warming_and_temperature_rise/global_warming%20and%20temperature%20rise/review/review_table.json

html :

https://github.com/semanticClimate/semantic_corpus/blob/shreya/corpora/shreya_project/templates/queries/global_warming_temperature_rise/review_table.html

md :

https://github.com/semanticClimate/semantic_corpus/blob/shreya/corpora/shreya_project/global_warming_and_temperature_rise/global_warming%20and%20temperature%20rise/review/review_table.md

VII. TOOLS AND TECHNOLOGIES USED

Python

During the internship, Python scripts were used to support literature processing, metadata handling, review table generation, and organization of the semantic corpus. Python also enabled automation

Git and GitHub

Git was used to monitor changes in project files, maintain a history of modifications, and facilitate collaborative development. The project repository was managed using GitHub, allowing organized maintenance of scripts, documentation, and corpus files. Version control ensured that updates to the corpus could be tracked systematically and that previous versions of the workflow could be recovered when required.

GitHub Codespaces and Visual Studio Code

The computational workflow was implemented using **GitHub Codespaces**, which provided a cloud-based development environment configured with the required software dependencies.

Visual Studio Code (VS Code) served as the integrated development environment (IDE) for editing Python scripts, managing project files, and executing workflow components. Features such as syntax highlighting, integrated terminal access, and source control integration improved productivity and facilitated efficient project management.

Slack

The slack was used as a platform for communication, progress updates and coordination with the team members

JSON (JavaScript Object Notation)

JSON was used as the primary structured data format for storing metadata and review information. Its lightweight syntax and hierarchical organization make it well suited for representing complex scientific datasets.

HTML-Based Review Viewer

An HTML review viewer was generated to improve the accessibility of the corpus. Rather than examining raw JSON files, users could inspect publication metadata through a structured web interface. The HTML review viewer demonstrated how computational visualization can enhance literature management without requiring specialized software.

Literature Repositories (Europe PMC)

Scientific publications included in the corpus were obtained through established literature repositories integrated into the workflow. These repositories provide access to peer-reviewed research articles and associated bibliographic metadata.

Markdown Documentation

Markdown files were used to document the workflow, project structure, and corpus organization. Their lightweight syntax facilitated clear documentation while remaining compatible with GitHub and other version control platforms.

VIII. CONCLUSION

The rapid expansion of scientific literature has made the organization and management of research information an increasingly important aspect of modern scientific investigation. Efficient retrieval and systematic organization of publications are essential for enabling evidence-based research, facilitating knowledge discovery, and ensuring reproducibility. The present internship addressed these challenges through the development of a semantic corpus for the research query **"Global Warming and Temperature Rise."**

A major outcome of the internship was the implementation of a reproducible workflow for organizing scientific literature. The use of standardized directory structures, consistent metadata formats, review tables, and documented procedures ensured transparency and facilitated efficient management of retrieved publications. These practices align with contemporary principles of research data management and contribute to the long-term usability of the corpus. It also demonstrated the value of integrating computational tools into scientific literature management. Programming with Python, version control through Git and GitHub, structured data storage using JSON, and visualization through an HTML-based review interface collectively enhanced the efficiency and reproducibility of the workflow. The experience highlighted how computational approaches can significantly improve the organization and accessibility of scientific information.

From a scientific perspective, the semantic corpus developed during this internship has potential applications beyond the immediate project. The structured dataset can support systematic literature reviews, bibliometric analyses, natural language processing, semantic search, and knowledge graph construction. Furthermore, the workflow is adaptable to a wide range of research topics and can serve as a general framework for developing semantic corpora in other scientific domains.

Overall, this internship has provided a comprehensive understanding of semantic corpus development and demonstrated the practical importance of reproducible research workflows in managing the growing volume of scientific literature. The knowledge and skills acquired through this work will serve as a strong foundation for future research in bioinformatics, computational biology, environmental informatics, and data science.

IX. REFERENCES

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4. Python Software Foundation. <https://www.python.org/>
5. Pandas Documentation. <https://pandas.pydata.org/>
6. Jupyter Notebook Documentation. <https://jupyter.org/>
7. Europe PMC REST API Documentation. <https://europepmc.org/RestfulWebService>
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